

Vanika Sok

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Education

Lafayette College

Bachelor of Arts in Mathematics
Bachelor of Arts in Economics

Easton, PA, USA
Aug 2022 – May 2026

Graduated with Honors in Economics, Cum Laude

Primary advisor: Dr. David Stifel
Second reader: Dr. Susan L. Averett
Third reader: Dr. Daniel Turek

Relevant Coursework

Real Analysis, Discrete Structure, Probability, Mathematical Statistics, Theoretical Math, Abstract Algebra, Advanced Regression Analysis, Linear Algebra, Stochastic Processes
Fundamentals of Econometrics, Microeconomics, Macroeconomics, Financial Market, Development Economics, Industrial Organization, Independent Study on Advanced Micro Development Economics, Women and the Economy, Thesis

Extracurricular

Orientation Leader and First-Year Mentor; Mentor for Lafayette Initiative for Malagasy Education (LIME)
Lafayette Investment Club, International Student Association, Women in Computing, Women in Math

Teaching Experience

Teaching Assistant, Lafayette College

ECON 253: Econometrics; Professor Matthew Larsen Fall 2025
PHYS 131: Mechanics; Professor Paul Stonaha Spring 2025
ECON 251: Intermediate Microeconomics; Professor Sayorn Chin Spring 2024

Honors and Awards

Omicron Delta Epsilon (ODE), the International Honor Society for Economics Apr 2025
Founding Chapter Leader, National Student Data Corps (NSDC), Lafayette College–Northeast Sep 2023
Big Data Innovation Hub

Best Overall Hackathon Award

Feb 2023

Best overall hackathon award from Lafayette College's Association for Computer Machinery for a project on a mental health website featuring a chatbox, games, and videos designed to alleviate stress.

SHE-CAN Scholar

Jan 2020 – Present

Selected for the SHE-CAN scholarship program based on excellence in academics, project management, leadership skills, and public speaking.

Professional Experience

United Nations Office for Project Services (UNOPS)

New York, NY

Digital & Innovation Strategy Intern (AI, GIS & IM)

May 2026 – Present

Support UNOPS PSC's Information Management and Analytics Team (IMAT), delivering data-driven solutions to mitigate explosive hazards and support crisis response across conflict-affected contexts. Assist in producing communication materials on AI-driven tools including CARE (prioritization) and ORBIT (orbital intelligence); support documentation of AI model interpretability for machine learning terrain modeling identifying explosive ordnance risks. Assist in preparing reports and visual presentations using geospatial observation data for damage assessments and environmental mapping; support organization and cataloging of high-resolution satellite imagery for 3D modeling and rubble estimation in reconstruction zones.

Research Experience

University of Utah, Office of Undergraduate Research

Salt Lake City, UT

Researcher, Dept. of Economics

May 2025 – Aug 2025

Engineered end-to-end data pipelines via web scraping and RESTful APIs to collect, clean, and merge 15+ years of large-scale panel data across the Public Library Survey, Bureau of Labor Statistics, and Business Dynamic Statistics using Python, R, and Stata. Applied OLS, fixed-effects, and instrumental variable models to produce the first causal estimates of unemployment's effect on public library usage across the U.S. (2008–2022). Co-authored two research papers in progress—one identifying that COVID-era unemployment reversed pre-pandemic library usage patterns, and one demonstrating that counties with higher single-mother rates and childcare costs show significantly greater children's programming attendance

—with findings directly informing resource allocation policy during economic downturns including the Great Recession and COVID-19.

Lafayette College, Dept. of Economics
Research Assistant

Easton, PA

May 2023 – May 2025

Applied two-way fixed-effects models to NLSY97 and O*NET data to quantify a statistically significant wage premium for morally compromising occupations, with premium magnitude increasing with age and education. Presented findings at Lafayette College's Poster Symposium, demonstrating how markets price moral considerations and offering insights with implications for workforce management, compensation strategies, and organizational decision-making.

Analyzed how crop diversification affects household welfare in Nigeria, leveraging World Bank panel survey data. Applied econometric models, data wrangling, and visualization in Excel and Stata to quantify the relationship between agricultural diversity and economic well-being. Produced insights on how diversification strategies can strengthen household resilience and inform development policy.

Research
Output

Honors Thesis

“Does Women’s Empowerment Improve Child Welfare? New Evidence from Cambodia”

Abstract Women’s empowerment in developing countries remains understudied, and in particular its impact on child welfare. I examine how women’s intra-household bargaining power – proxied by women’s land ownership, education, perception about justified wife-beating, and household decision-making – affects child welfare indicators. Using Demographic and Health Survey (DHS) data from Cambodia from 2005 to 2022, I estimate how women’s empowerment affects child school enrollment status, immunization rates, and child anthropometric health outcomes. I also explore whether these impacts differ by the gender of the child, and potentially reduce any gender-based inequality in child welfare. In order to reduce the bias in the estimators of these effects, I control for household level characteristics and region and time fixed effects. The findings will contribute to the literature on gender and development by providing new evidence from Cambodia and offering insights for policies targeting child welfare.

Works in Progress

“Further Evidence on Pollution, Crime, and Anti-Social Behavior” coauthored with Sayorn Chin and Salvador Lurbé (Lead Data Scientist at Mountain Data Group, Fort Collins, Colorado)

Abstract Recent literature has identified an association between short-term pollution exposure and crime in the US and in the United Kingdom (UK). In this paper, we expand our understanding of the effects of pollution on crime by looking at the association between pollution, specifically at Particulate Matter (PM) and Ozone (O₃), and counts of aggregated crime types and Anti-social Behavior (ASB) in the UK. We primarily focus our analysis on ASB, as the literature has identified costs associated with them in the UK, but has overlooked its association with pollution. Using a fixed effects model, we found an association between pollution and some crime types, especially those that are economically motivated. We find weaker evidence of an association between pollution and offenses associated with aggressive and violent behavior (including ASB), and we discuss potential mechanisms in the context of the rational choice crime model. We conclude that one potential mechanism could be a decrease in the utility of the non-punishable alternative activity, or a decrease of the offender’s value of the future costs associated with being caught.

“Unemployment and Public Library Usage in the United States: Evidence from Before and During the COVID-19 Pandemic” coauthored with Sarah Small and Rachel Ernst (Department of Economics and J. Willard Marriott Library, University of Utah, Salt Lake City, Utah)

Abstract Scholars and policymakers have long argued that public libraries offer essential community resources during times of economic strife. However, to our knowledge, most of this evidence has been anecdotal or correlational. Using Public Library Survey, Bureau of Labor Statistics, and Business Dynamic Statistics datasets, we present causal estimates of the effect of regional unemployment on local public library usage. From OLS and instrumental variable models, we find that from 2008 to 2019, increased rates of unemployment led to increased visits, circulation, and computer usage at local libraries. However, during the COVID-19 pandemic, many of these relationships reversed, with COVID-related unemployment causing a reduction in visits, but not necessarily use. Reflecting on various dimensions of community need and library resources, we contextualize our results in the broader literature on public libraries as valuable economic resources.

“Do Public Libraries Fill Care Gaps? Regional Analyses of Children’s Library Programming, Childcare Markets, and Parenthood” coauthored with Sarah Small and Rachel Ernst

Abstract Public libraries are regularly relied upon for community resources beyond just books. In this paper, we ask: do public libraries help fill care gaps? Using data from the United States Public Library Surveys and the National Database on Childcare Prices, we construct a panel of public libraries from 2008 to 2022 matched with corresponding demographic and childcare pricing information in their respective counties. Using these data, we estimate how family demographics and care costs relate to attendance in children’s programming and circulation of children’s books in public libraries. Ultimately, we find that in counties with more single mothers, higher home-based childcare prices, and higher rates of family-level poverty, libraries have higher attendance in young children’s programming activities and greater children’s book circulation. We test the robustness of our results using several instrumental variable analyses and falsification tests. Ultimately, we suggest that libraries serve as important public spaces of social reproduction.

Research
Presentation

Poster Symposiums and Presentations

Presented research poster “Unemployment and Public Library Usage in the United States” at Lafayette College Symposium (Fall 2025) and Lafayette Bicentennial Celebration.

Presented research poster “The Compensation of Conscience” at Lafayette College Poster Symposium (Spring 2025).

Presented research poster “Unemployment and Public Library Usage in the United States” at University of Utah Undergraduate Research Symposium (Summer 2025).

Shared findings on childcare markets and public library programming at University of Utah Summer Research Symposium (2025).

Research
Assistant

“The Compensation of Conscience”, Sayorn Chin

This paper has been submitted to the *Journal of Economic Behavior and Organization*

Abstract Markets are often praised for their efficiency, but they can also incentivize individuals to act against their moral convictions. This study quantifies the “price of conscience” in labor markets, investigating whether jobs requiring moral compromise offer higher wages. Using data from the National Longitudinal Survey of Youth 1997 (NLSY97) and Occupational Information Network (O*NET), we conduct a longitudinal fixed-effects analysis to examine how conscience-related job characteristics influence wages. Our findings suggest that occupations demanding greater moral compromise provide higher wages, and this premium increases with age and education. These results extend prior experimental evidence on morals and markets into real-world labor market dynamics.

Term
Projects

Department of Mathematics, Lafayette College

Probabilistic Forecast Calibration and Model Validation Study of Major League Baseball
Regression-Based Time-Series Enrollment Forecasting for Lafayette Spring 2026 Math Courses
Proportional-Odds Modeling of Conspiracy Theory Beliefs Using Information-Criterion-Based Selection

Department of Economics, Lafayette College

Analyzing JetBlue’s Strategy: From Levitt’s Theory to Limited Information and Pricing
Incomplete Contracts and Market Breakdown in High-Risk Homeowners Insurance
Does Rigorous High School Exposure to STEM Education Improve Long-Term Employment Outcomes?
Lafayette Dining Optimization: Utility and Risk Modeling

Professional
Certificate

Training the Street

Financial Modeling and Corporate Valuation

New York City, New York
Mar, 2024

IBM

Data Science Specialization

(Online)
May, 2024

Skills

Programming

Python, R, MySQL

Methods

Causal Inference, Difference-in-Differences, A/B Testing, Regression Discontinuity, Instrumental Variables, Randomized Controlled Trials, Fixed Effects Models, OLS, Panel Data Analysis, Web Scraping, RESTful APIs, Data Pipeline Engineering

Software

Git/GitHub, Stata, $\text{\LaTeX}$, Tableau, Microsoft Excel

Languages

English (fluent), Khmer (native), Chinese (proficient), French (conversational)

Conferences

The National Conference on Undergraduate Research (NCUR), 2026 Richmond, VA
Does Women's Empowerment Improve Child Welfare? New Evidence from Cambodia. Oral presenter and session moderator for undergraduate student presenters.

Economic Scholars Program, Federal Reserve Bank of Cleveland, 2026 Cleveland, OH
Federal Reserve-hosted economic research conference for students to share research, network with professionals, and explore Fed career pathways.

UN International Day of Women and Girls in Science Celebration, 2024 New York City, NY
Represented Lafayette College's Inclusive STEM Education initiatives at the United Nations headquarters. Participated in high-level UN sessions on gender equity in STEM education and workforce participation. Applied policy insights to strengthen LIME mentoring initiatives, Women in Math, and Women in Computing outreach programming on Lafayette campus.

Grace Hopper Celebration (GHC), 2023 Orlando, FL
Represented Lafayette College's Women in Computing at the Grace Hopper Celebration, the world's largest conference for women in technology. Participated in technical sessions, industry panels, and the career fair; engaged with recruiters, engineers, and researchers on emerging trends in AI, data science, cybersecurity, and quantitative computing.

SHE-CAN Leadership Summit 2024 and 2026 San Francisco, CA
Annual, weekend long, in-person summit in the San Francisco Bay Area. Summit Mission Statement: The "Perpetual Power of One" is the realization that individual actions and contributions, no matter how small they may seem at the outset, can create a ripple effect that continues over time and inspires others to make their own contributions.

Leadership Summit, 2023 San Francisco, CA
Selected to attend the 2023 SHE-CAN Leadership Summit, an annual weekend-long event in the San Francisco Bay Area, focused on how leadership can drive sustainability, conscious capitalism, social justice, and the empowerment of women.

Leadership and Volunteering

Lafayette Data Community Easton, PA
Board Member Apr 2023 – Present
Coordinate campus-wide data science programming, host technical workshops, and invite speakers on AI, quantum computing, and computational sustainability. Initiate and lead Inspire-a-Thon with Hanson Center, a campus-wide AI-focused innovation challenge featuring multidisciplinary project tracks in policy, technology, and creative expression.

Lafayette Extended Orientation (LEO) Program Easton, PA
LEO Leader and First-Year Student Mentor Apr 2025 – Present
Provides support and mentorship to new students Class of 2029 during their first year as they navigate their new intellectual community both socially and academically.

Lafayette Initiative for Malagasy Education (LIME) Madagascar
Math Tutor Dec 2024 – Present
Selected for a 1.5-year commitment with the LIME; traveled to Antananarivo, Madagascar (Winter 2025) to mentor high school students, providing academic guidance, leadership training, and support in preparing for competitive university and scholarship applications to the U.S.

Lafayette Intercultural Development Easton, PA
International Orientation Captain May 2023 – May 2024

Led the planning and execution of a week-long International Orientation program serving students from 52 countries in the Classes of 2027 and 2028.

Cambodian Student Association in the United States

Washington, DC (Remote)

Director of Finance

Sep 2023 – May 2025

Managed all financial operations, securing grants from the Cambodian Embassy and extending the student budget by 125 percent. Oversaw fund allocation across departments to maximize ROI, maintained full financial records, and collaborated with the Board of Directors to authorize and disburse funds in compliance with CASUSA's constitution, supporting the association's mission to build a welcoming community for Cambodian students in the U.S.

SHE-CAN Make A Change Project: Higher Education Fair

Siem Reap, Cambodia

Founder

Mar - Jun 2024

Brought the first-ever Higher Education Fair to live at a Rural High School in Siem Reap province. Drawn over 300 students participation from 12 different high school across Cambodia and invited four incredible speakers. All this made possible due to the help of my younger sister, and a team of six brilliant volunteers whom we recruited from 200+ student who filled out our volunteer application form.

Center of Khmer Studies and American Corner

Siem Reap, Cambodia

1st and 2nd Book Fair Team Leader and Speaker

Apr – Jul 2022 and May – Jun 2023

Planned and executed 3-month project to organize inaugural book fair in Siem Reap. Sent invitations to 20 high schools, universities, and other educational institutions, driving 5,000 in-person attendees. Collaborated with 25 publishers, libraries, bookstores, universities, and educational institutions to display their products, securing donations, increasing event attendance 525 percent over projections. Supervised Booth and Registration teams and emceed 5000 person event. After the great success of the first book fair, our team returned in 2023 with more major exhibitors from across the country.

United States Agency for International Development

Siem Reap, Cambodia

Student Representative

Nov 2021

Led group discussions on career development and education challenges in rural areas Cambodia, delivering public speeches and fostering group discussions on 5-year plans to develop Cambodia.

Cambodia Women's Crisis Center

Siem Reap, Cambodia

Community Service Volunteer

Aug – Oct 2021

Organized events for 35 families in my community to raise awareness of domestic violence and sex trafficking. Organized events with the center to raise awareness and educate community and incoming tourists. Directed underprivileged households to the hotlines number to reduce impact of domestic assault across Cambodia.

References

David Stifel (Advisor)

Professor, Department of Economics

Department Head

Lafayette College

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Sayorn Chin

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Sarah Small

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